

Decent Coaching Classes
X SB
Academic Conquest - VI

Subject : Mathematics (Part I – Full Book ; Part II – Ch. 1, 2, 5, 6)

Marks : 30

Q.1 (A) Choose the correct alternative for each sub-question.

[4]

- (1) The rate of GST on an article is 18%. The rate of CGST on it is :
(A) 18% (B) 9% (C) 36% (D) 4.5%
- (2) A die is rolled. The probability of getting a prime number on the upper face is :
(A) $1/6$ (B) $1/3$ (C) $1/2$ (D) $2/3$
- (3) Which of the following is a Pythagorean triplet?
(A) (6, 8, 12) (B) (5, 12, 13) (C) (7, 9, 12) (D) (9, 12, 16)
- (4) $1 + \tan^2\theta = ?$
(A) $\cot^2\theta$ (B) $\operatorname{cosec}^2\theta$ (C) $\sec^2\theta$ (D) $\tan^2\theta$

Q.1 (B) Solve the following sub-questions.

[4]

- (1) Find the value of the discriminant of the equation $2x^2 - 5x + 2 = 0$.
- (2) Find the 10th term of the A.P. 3, 7, 11, 15, ...
- (3) Find the distance between the points A(2, 3) and B(4, 1).
- (4) The ratio of the corresponding sides of two similar triangles is 3 : 5. Find the ratio of their areas.

Q.2 (A) Complete any TWO of the following activities.

[4]

- (1) Solve the simultaneous equations : $2x + y = 7 \dots (I)$; $x - y = 2 \dots (II)$
Adding (I) and (II) : $3x = \underline{\hspace{2cm}}$ $\therefore x = \underline{\hspace{2cm}}$
Putting $x = 3$ in (I) : $2 \times \underline{\hspace{2cm}} + y = 7 \therefore y = \underline{\hspace{2cm}}$
 $\therefore$ Solution is $(x, y) = (\underline{\hspace{2cm}}, \underline{\hspace{2cm}})$
- (2) In ΔPQR , $\angle Q = 90^\circ$ and seg $QS \perp$ seg PR . If $PS = 9$ and $SR = 4$, find QS .
By the theorem of geometric mean, $QS^2 = PS \times \underline{\hspace{2cm}}$
 $\therefore QS^2 = \underline{\hspace{2cm}} \times \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$
 $\therefore QS = \underline{\hspace{2cm}}$
- (3) If $\tan\theta = 3/4$, find $\sec\theta$ and $\cos\theta$.
 $\sec^2\theta = 1 + \underline{\hspace{2cm}} = 1 + (3/4)^2 = 1 + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$
 $\therefore \sec\theta = \underline{\hspace{2cm}}$, $\cos\theta = 1 / \sec\theta = \underline{\hspace{2cm}}$

Q.2 (B) Solve any TWO of the following.

[4]

- (1) If $x = 3$ is a root of the equation $kx^2 - 10x + 3 = 0$, find the value of k .
- (2) The taxable value of a wrist-watch belt is ₹ 586 and the rate of GST is 18%. Find the price of the belt for the customer.
- (3) Find the coordinates of the midpoint of the segment joining the points (22, 20) and (0, 16).

Q.3 (A) Complete any ONE of the following activities.

[3]

- (1) Two coins are tossed. Find the probability of getting at least one head.
 $S = \{ HH, HT, TH, TT \} \therefore n(S) = \underline{\hspace{2cm}}$
Event A : at least one head $\therefore A = \{ \underline{\hspace{2cm}}, \underline{\hspace{2cm}}, \underline{\hspace{2cm}} \} \therefore n(A) = \underline{\hspace{2cm}}$
 $P(A) = n(A) / n(S) = \underline{\hspace{2cm}} / \underline{\hspace{2cm}}$
- (2) The areas of two similar triangles are 225 sq. cm and 81 sq. cm. If a side of the smaller triangle is 12 cm, find the corresponding side of the bigger triangle.
 $A(\text{smaller}) / A(\text{bigger}) = (12 / x)^2 \therefore \underline{\hspace{2cm}} / \underline{\hspace{2cm}} = 144 / x^2$

$$\therefore 12 / x = \underline{\hspace{1cm}} / \underline{\hspace{1cm}} \quad \therefore x = \underline{\hspace{1cm}} \text{ cm}$$

Q.3 (B) Solve any TWO of the following.

[6]

- (1) How many three-digit natural numbers are divisible by 5?
- (2) Find the mean of the following distribution by the direct method.

Milk sold (litres)	1 – 2	2 – 3	3 – 4	4 – 5	5 – 6
No. of customers	17	13	10	7	3

- (3) Prove that : $(\sec\theta + \tan\theta)(1 - \sin\theta) = \cos\theta$

Q.4 Solve any ONE of the following.

[5]

- (1) The sum of a father's age and twice the age of his son is 70. If the father's age is doubled and added to the son's age, the sum is 95. Find their present ages.
- (2) From the top of a lighthouse 90 m high, the angle of depression of a ship is 60° . Find how far the ship is from the lighthouse. ($\sqrt{3} = 1.73$)
- (3) Mr. Amol purchased 50 shares of face value ₹ 100 when the market value of a share was ₹ 80. The company paid a dividend of 20%. Find (i) his total investment, (ii) the total dividend received, (iii) his rate of return on the investment.